Problem 6 | Problems T5_MechanicalProperties.pdf #6 | Hooke's law · design

Maximum Length of a Ni Alloy Sample for a Given Allowed Elongation

Original statement · Problems T5_MechanicalProperties.pdf, p. 3

6. A sample made of a Ni alloy with an elastic modulus of 207 GPa and an initial diameter of 10.2 mm experiences an elastic deformation only when loads inferior to 8900 N are applied. Calculate the maximum length of the sample before the deformation if the maximum elongation allowed is 0.25 mm.

Official Solution: $l_0 = 475$ mm

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Elastic up to 8900 N.** For loads below 8900 N the deformation is only elastic, so Hooke's law $\sigma = E\varepsilon$ holds [Session 11 Slide 13]. The load of 8900 N is therefore the largest load for which the sample is still elastic.
2. **Uniaxial, uniform stress** along the sample: $\sigma = F/A_0$ and $\varepsilon = \Delta l/l_0$ [Session 11 Slide 9].
3. **Worst case.** The elongation grows with the load and with the length. The longest sample that does not exceed the allowed elongation is found for the largest load that is applied, 8900 N.
4. The cross-section is a circle of 10.2 mm, and it does not change appreciably in the elastic range.

Data

Quantity	Symbol	Value	In SI
Young's modulus	E	207 GPa	207 000 N/mm²
Diameter	d_0	10.2 mm	10.2×10^{-3} m
Largest elastic load	F	8900 N	8900 N
Maximum elongation allowed	Δl_{max}	0.25 mm	0.25×10^{-3} m

Unknown: the maximum initial length l_0.

PHASE 2 · FRAMES & STRATEGY

One direction (the axis of the sample), no change of basis.

Strategy. Hooke's law relates the stress produced by the load to the strain, and the strain relates the elongation to the length. Combining the three relations,

$$\sigma = \frac{F}{A_0} \qquad \varepsilon = \frac{\sigma}{E} \qquad \Delta l = \varepsilon\, l_0$$

the elongation is proportional to the initial length. The condition $\Delta l \le 0.25$ mm is then an upper bound for l_0, and the maximum length is the one for which $\Delta l = 0.25$ mm.

PHASE 3 · STEP-BY-STEP DERIVATION

1. Area

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(10.2\ \text{mm})^2}{4}$$

The square is $10.2^2 = 104.04$, then $\pi \times 104.04 = 326.85$ and dividing by 4:

$$A_0 = 81.71\ \text{mm}^2$$

2. Stress and strain at the largest elastic load

$$\sigma = \frac{F}{A_0} = \frac{8900\ \text{N}}{81.71\ \text{mm}^2} = 108.92\ \text{MPa}$$

$$\varepsilon = \frac{\sigma}{E} = \frac{108.92\ \text{N/mm}^2}{207000\ \text{N/mm}^2} = 5.2617 \times 10^{-4}$$

3. Maximum initial length

From $\Delta l = \varepsilon\, l_0$ with $\Delta l = 0.25$ mm:

$$l_0 = \frac{\Delta l}{\varepsilon} = \frac{0.25\ \text{mm}}{5.2617 \times 10^{-4}} = \mathbf{475.1\ mm}$$

The same result in one step, without the intermediate strain:

$$l_0 = \frac{\Delta l\, A_0\, E}{F} = \frac{0.25\ \text{mm} \times 81.71\ \text{mm}^2 \times 207000\ \text{N/mm}^2}{8900\ \text{N}} = 475.1\ \text{mm}$$

PHASE 4 · CHECKS & INTERPRETATION

Checks

- **Units.** $\Delta l\, A_0 E / F$: mm · mm² · (N/mm²) / N = mm.
- **Back substitution.** For $l_0 = 475.1$ mm, $\Delta l = 5.2617 \times 10^{-4} \times 475.1 = 0.250$ mm, the allowed elongation.
- **Elastic range.** The strain, 5.2617×10^{-4}, is of the order of 10^{-4}, typical of the elastic deformation of a metal, so the use of Hooke's law is consistent with the statement.
- **Limit.** For a shorter sample the elongation is smaller (it is proportional to l_0) and the condition is satisfied with margin; for a longer one it is exceeded.

Interpretation

The sample can be at most about 475 mm long. The limit is imposed by the stiffness E (a stiff material elongates little) and by the section (a thicker sample has less stress for the same load); it is not imposed by the strength, because the stress stays elastic.

Check against the official solution

Item	Official	Computed	Comment
l_0	475 mm	475.1 mm	agrees